

Prime-Shell Hilbert Lift and the Sharp Collapse Barrier

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August 2026

Abstract

The previous structural stage attached a literal common-source twin-prime kernel to a finite interval by a reduced rational-frequency large sieve, but it collapsed the prime shell before the finite-window estimate. We retain the prime label and the four-packet label as Hilbert coordinates. For $H = x^{21/32}$, $Q = x^{1/3}$, and $U = x^{133/400}$, fixed- q cutoff injectivity and an elementary unsigned cluster bound give

$$\frac{1}{N} \sum_{n \in I_x} \|\mathbf{K}(n)\|_2^2 \ll JM^2 x^{1/96} (\log x)^5.$$

Here $\mathbf{K}$ has coordinates (j, q) , J is the fixed packet count, and M is a uniform profile bound. Recombining the prime labels costs exactly the elementary factor $P = \#\mathcal{Q}_x \leq 2Q$ and recovers the earlier scalar power $x^{11/32}$. A finite aligned fixture attains the full ratio $P = 4$, while a parallel four-packet algebraic fixture has unit-projection ratio one. Thus the new theorem preserves the interfaces and isolates the precise prime-shell collapse barrier; it supplies no arithmetic cancellation, L^2 advance, fixed-atom credit, or twin-prime conclusion.

Claim level. `PROVED_STRUCTURAL_L1 / PRIME_LABEL_AND_PACKET_PRESERVING_LIFT.`

The Hilbert lift, exponent ledger, and scalar recovery are proved under the declared common-source hypotheses. The finite fixtures are exact structural controls, not asymptotic arithmetic evidence.

1 Introduction

A useful analytic route toward the twin-prime problem should expose every place where a collective estimate is required. The recent finite-window attachment did this for a common-source kernel by regrouping divisor rows according to their reduced rational frequencies and applying the additive large sieve. That argument was intentionally unsigned. In particular, it collapsed the prime shell before invoking the finite-window inequality.

The present paper asks a narrower follow-up question: can the outer prime label and the four-packet label be retained in the same finite-window object? This is important for bookkeeping. If the labels are removed too early, a later signed prime-shell or polarized reassembly can be accidentally replaced by a Cauchy estimate without recording its cost. If they are retained, one can measure exactly what the structural argument does and does not provide.

We work with the same V46 common-source scales

$$H = x^{21/32}, \quad Q = x^{1/3}, \quad Y_0 = \frac{H}{4Q}, \quad U = x^{133/400}.$$

The prime shell is

$$\mathcal{Q}_x = \{q \text{ prime} : Q < q \leq 2Q\}, \quad P = \#\mathcal{Q}_x,$$

and the squarefree transition band is

$$\mathcal{D}_x = \{d \in \mathbb{Z} : Y_0 < d \leq U, \mu(d)^2 = 1\}.$$

The physical interval is $I_x = (x/2, x] \cap \mathbb{Z}$, with $N = |I_x|$.

Our main result is a tensorized version of the standard additive large sieve. It preserves (q, j) as coordinates and gives a normalized split exponent $1/96$, since

$$\frac{Q^2}{H} = x^{1/96}.$$

Only after this result do we form the scalar packet shell. Pointwise Cauchy over q costs $P \leq 2Q$, changing Q^2/H into $Q^3/H = x^{11/32}$. This identity is the central accounting statement of the paper.

The contributions are:

1. an exact prime-label- and packet-label-preserving finite-window Hilbert lift;
2. an explicit split-versus-scalar exponent ledger, with the P cost isolated rather than hidden;
3. a positive-semidefinite packet Gram formulation compatible with exact four-point complex polarization;
4. two scoped adversarial controls: a P -saturating prime alignment and a geometry-only packet alignment.

Status boundary. Nothing here proves a signed estimate for the literal prime shell. In the terminology of the route ledger,

$$\text{ARITHMETIC_ADVANCE=NO,} \quad \text{FIXED_ATOM_CREDIT=0,} \quad \text{L2=NONE.}$$

The result is a structural bridge to the next open theorem.

2 The split common-source object

Let $\psi_j : \mathbb{R} \rightarrow \mathbb{C}$ be bounded profiles for $0 \leq j < J$, and put

$$M = \max_{0 \leq j < J} \|\psi_j\|_\infty.$$

For $q \in \mathcal{Q}_x$, $h \leq U$, and $a \pmod{h}$, define

$$B_{h,q}^{(j)}(a) = \sum_{0 < |m| \leq \lfloor hq/H \rfloor} \psi_j\left(\frac{Hm}{hq}\right) \mathbf{1}_{m\bar{q} \equiv a \pmod{h}}. \quad (1)$$

The row is defined on all residues; only primitive residues will be used as reduced-frequency numerators. Set

$$c_d = \frac{\mu(d) \log d}{d}, \quad C_h = \sum_{\substack{d \in \mathcal{D}_x \\ h|d}} c_d. \quad (2)$$

The split coordinates are

$$K_{j,q}(n) = \sum_{h \leq U} \sum_{\substack{a \bmod h \\ (a,h)=1}} C_h B_{h,q}^{(j)}(a) e(na/h), \quad \mathbf{K}(n) = (K_{j,q}(n))_{\substack{0 \leq j < J \\ q \in \mathcal{Q}_x}}. \quad (3)$$

The scalar packet shell is

$$K_j(n) = \sum_{q \in \mathcal{Q}_x} K_{j,q}(n). \quad (4)$$

The exponent relations needed below are

$$\frac{Q^2}{H} = x^{1/96}, \quad \frac{Q^3}{H} = x^{11/32}, \quad \frac{U^2}{x} = x^{-67/200}, \quad \frac{UQ}{H} = x^{23/2400}. \quad (5)$$

For sufficiently large x , $4Q < H$ and $U < Q$. In particular every shell prime is a unit modulo every divisor in $\mathcal{D}_x$.

2.1 The inherited reduced-frequency identity

The divisor-dilation identity from the preceding finite-window stage is

$$B_{d,q}^{(j)}(ka) = B_{h,q}^{(j)}(a) \quad (d = kh, \ h \mid d), \quad (6)$$

because the congruence forces $m = kn$ and the cutoff and profile argument scale exactly. Thus the literal common-source kernel can be partitioned by the reduced frequency a/h while retaining (q, j) . This is an identity, not an orthogonality assertion.

3 The Hilbert-valued finite-window theorem

Lemma 3.1 (fixed- q cutoff injectivity). *For sufficiently large x ,*

$$\sum_{\substack{a \bmod h \\ (a,h)=1}} \left| B_{h,q}^{(j)}(a) \right|^2 \leq 2M^2 \frac{hq}{H}.$$

Proof. If two nonzero atoms m_1, m_2 in (1) occupy the same residue, then $h \mid (m_1 - m_2)$. Since $q \leq 2Q$ and $4Q < H$,

$$|m_1 - m_2| \leq 2 \left\lfloor \frac{hq}{H} \right\rfloor < h.$$

Thus the atoms are distinct modulo h . Summing their squared magnitudes over all residues, and then restricting to primitive residues, gives at most $2 \lfloor hq/H \rfloor M^2 \leq 2M^2 hq/H$. $\square$

Lemma 3.2 (active-cluster harmonic bound). *The literal coefficients satisfy*

$$\sum_{h \leq U} h |C_h|^2 \ll (\log x)^5.$$

Proof. If the row indexed by h is nonzero for at least one shell prime, then $\lfloor hq/H \rfloor \geq 1$, hence $h \geq H/(2Q)$. Writing $d = hk$ in (2), we have $k \leq U/h \leq 2UQ/H$. Therefore

$$|C_h| \leq \frac{\log U}{h} \sum_{k \leq U/h} \frac{1}{k} \ll \frac{(\log x)^2}{h}.$$

The active h form a subinterval of $[H/(2Q), U]$, so

$$\sum_h h|C_h|^2 \ll (\log x)^4 \sum_{H/(2Q) \leq h \leq U} \frac{1}{h} \ll (\log x)^5.$$

No cancellation in μ is used. $\square$

Lemma 3.3 (Hilbert-valued additive large sieve). *Let I be a consecutive interval of N integers. Then*

$$\sum_{n \in I} \|\mathbf{K}(n)\|_2^2 \leq (N + U^2) \sum_{\substack{h \leq U \\ (a, h) = 1}} \sum_{q \in \mathcal{Q}_x} \sum_{0 \leq j < J} |C_h B_{h,q}^{(j)}(a)|^2.$$

Proof. Distinct reduced fractions with denominators at most U have circular spacing at least U^{-2} . Apply the standard additive large-sieve inequality [1] to the scalar expansion of each coordinate $K_{j,q}(n)$, with coefficient $C_h B_{h,q}^{(j)}(a)$, and sum the resulting inequalities over (q, j) . This is precisely the coordinate realization of the tensor lift from scalar coefficients to $\ell^2(\mathcal{Q}_x \times \{0, \dots, J-1\})$. $\square$

Theorem 3.4 (prime-label- and packet-preserving lift). *For fixed J and bounded packet profiles,*

$$\sum_{n \in I_x} \|\mathbf{K}(n)\|_2^2 \ll JM^2(N + U^2) \frac{Q^2}{H} (\log x)^5.$$

Consequently,

$$\frac{1}{N} \sum_{n \in I_x} \|\mathbf{K}(n)\|_2^2 \ll JM^2 x^{1/96} (\log x)^5,$$

and the unnormalized exponent is $97/96 + o(1)$.

Proof. By Lemma 3.1,

$$\begin{aligned} & \sum_{h,a,q,j} |C_h B_{h,q}^{(j)}(a)|^2 \\ & \leq \frac{2M^2 J}{H} \left(\sum_{q \in \mathcal{Q}_x} q \right) \sum_h h|C_h|^2 \\ & \ll JM^2 \frac{PQ}{H} (\log x)^5 \ll JM^2 \frac{Q^2}{H} (\log x)^5, \end{aligned}$$

where $P \leq 2Q$. Insert this into Lemma 3.3. Since $U^2/N = x^{-67/200+o(1)}$, division by N gives the normalized estimate. $\square$

3.1 Scalar recovery and the packet Gram

Corollary 3.5 (exact scalar collapse ledger). *For the packet shell (4),*

$$\frac{1}{N} \sum_{n \in I_x} \sum_{j < J} |K_j(n)|^2 \ll JM^2 x^{11/32} (\log x)^5. \quad (7)$$

Proof. For each j, n , Cauchy gives $|\sum_q K_{j,q}(n)|^2 \leq P \sum_q |K_{j,q}(n)|^2$. Sum over j, n and apply Theorem 3.4; then use $P \leq 2Q$ and $Q^3/H = x^{11/32}$. $\square$

For a finite interval define the packet Gram matrix

$$G_I[j, \ell] = \sum_{n \in I} K_j(n) \overline{K_\ell(n)}. \quad (8)$$

It is positive semidefinite, and Corollary 3.5 is a trace bound. For every unit packet vector ω ,

$$\sum_{n \in I} \left| \sum_j \overline{\omega_j} K_j(n) \right|^2 = \omega^* G_I \omega \leq \text{tr}(G_I). \quad (9)$$

The exact four-packet polarization interface remains available through

$$x\overline{y} = \frac{1}{4} \sum_{r=0}^3 i^r |x + i^r y|^2. \quad (10)$$

The trace inequality is unsigned and therefore does not supply the signed four-packet estimate needed by the terminal route.

4 Adversarial controls

4.1 A P-saturating prime-label fixture

Take

$$d = 5, \quad H = 500, \quad \{q\} = \{101, 131, 151, 181\}, \quad \psi(t) \equiv 1.$$

All four primes are 1 (mod 5), and $\lfloor dq/H \rfloor = 1$. Hence every fixed- q row is $e_1 + e_4$. The coherent row and diagonal energies are

$$\left\| \sum_q B_{5,q} \right\|_2^2 = 32, \quad \sum_q \|B_{5,q}\|_2^2 = 8,$$

so their ratio is $4 = P$. This is an exact rational finite structural adversary. It refutes only the scoped shortcut that fixed- q row geometry would imply a free or sub- P shell collapse.

4.2 A packet-projection alignment

Let $v \neq 0$ and let ω be a unit four-vector. Set $Z^{(j)} = \omega_j v$. Then

$$\sum_j \|Z^{(j)}\|^2 = \|v\|^2, \quad \left\| \sum_j \overline{\omega_j} Z^{(j)} \right\|^2 = \|v\|^2.$$

For $\omega = (1, i, -1, -i)/2$, the projection-to-total ratio is one. This is an algebraic Hilbert-space obstruction: packet geometry alone cannot create cancellation. It is not a claim that every literal TPC profile realizes this parallel family.

Table 1: TPC-218 claim firewall.

Structural lift	PROVED_STANDARD_TENSOR_LIFT
Split exponent	PROVED_1_OVER_96_LOG_FIVE
Scalar recovery	PROVED_P_FACTOR_RECOVERS_11_OVER_32
Prime-label orthogonality	REFUTED_SCOPED
Packet cancellation	NONE
Arithmetic advance	NO
Prime shell / four packets	OPEN
Full Gate B / strict one-over-400	OPEN / UNPAID

5 Route position and reproducibility

The result belongs to Route B’s structural threshold-A layer. Route A is not applicable. The strongest positive result is the label-preserving split envelope with exponent $1/96$, together with the exact identification of the single P factor needed for scalar recovery. The strongest obstruction is the finite P -saturating row alignment. The open theorem is a signed prime-shell reassembly that beats this cost while retaining the literal four-packet, zero/nonunit, fixed-atom, and normalization interfaces.

The repository release contains a producer, an independent checker, normal and optimized-mode checks, and a separate adversarial script. The finite dilation fixture uses the exact rational surrogate $\mu(d)/d$ solely to check the index map; the asymptotic theorem above uses the literal $\mu(d) \log(d)/d$. No numerical fixture is used as asymptotic evidence.

6 Conclusion

TPC-218 turns the instruction “keep the shell” into a precise Hilbert-valued theorem. The split object is controlled at the $Q^2/H = x^{1/96}$ scale, while scalar shell recovery necessarily records a $P \leq 2Q$ payment and returns the TPC-217 exponent $11/32$. The finite aligned row and the parallel packet construction show why geometry alone cannot pay the missing arithmetic saving. The next research target is therefore unambiguous:

prove a signed prime-shell reassembly beyond the exact P collapse.

This paper makes no claim that the target has been achieved.

Acknowledgement of scope. The claims are limited to the displayed common-source object and its declared finite-window normalization. In particular, `TPC218_ARITHMETIC_ADVANCE=NO`, `TPC218_FIXED_ATOM_CREDIT=0`, and `TPC218_FULL_GATE_B=OPEN`.

References

- [1] Hugh L. Montgomery and Robert C. Vaughan. *The large sieve*, volume 20. Mathematika, 1973.